ModelIt!

Teacher's Guide

Why Does Your Bike Rust?

How Water and Oxygen Change Metal Over Time

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-PS1-2, 4-ETS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-PS1-2, 4-ETS1-1

Measure and graph quantities to provide evidence that when substances are mixed, new substances may be produced. Define a simple design problem reflecting a need or want that includes specified criteria for success and constraints on materials, time, or cost.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Planning Investigations** — Students develop a five-component model of the rusting process and plan investigations to test protective coatings, controlling variables and measuring outcomes.
Disciplinary Core Idea	**PS1.B Chemical Reactions / ETS1.A Defining and Delimiting Engineering Problems** — When iron is exposed to water and oxygen, a chemical reaction produces a new substance (iron oxide/rust) with different properties. Students define the problem of preventing rust and test solutions.
Crosscutting Concept	**Cause and Effect** — Students identify causal relationships between environmental conditions (water, oxygen) and material changes (oxidation, rust, weakening), understanding that chemical reactions produce new substances with different properties.

CAST Testing Alignment

MULTIPLE CHOICE:

A student places four identical steel nails in different conditions: (A) in water with air, (B) in water with no air, (C) dry with air, (D) dry with no air. After two weeks, nail A has the most rust. Which best explains why?

CONSTRUCTED RESPONSE:

A bridge made of steel is starting to show rust. Using what you know about oxidation, water, and oxygen, explain why the bridge is rusting and recommend two different ways to slow or stop the rusting. Use the words 'chemical reaction,' 'oxidation,' and 'protective coating' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Rust is one of the most common and costly chemical reactions in the world. The United States alone spends over 270 billion dollars each year repairing and preventing rust damage to bridges, buildings, vehicles, and infrastructure. Understanding the chemistry of rust helps engineers protect the metal structures that hold up our modern world.

The Chemistry of Rust

Rust (iron oxide, Fe2O3) forms through a chemical reaction called oxidation. When iron (Fe) atoms on a metal surface come into contact with water (H2O) and oxygen (O2), they undergo a series of reactions. Water acts as a catalyst, helping oxygen molecules reach and react with iron atoms. The iron atoms lose electrons to the oxygen atoms, forming iron oxide — a completely new substance with entirely different properties. The original iron is shiny, strong, and magnetic. Rust is dull, weak, crumbly, and barely magnetic. This is a one-way chemical change — you cannot turn rust back into iron without extreme heat and chemical processing.

Why Both Water AND Oxygen Are Needed

Rusting requires both water and oxygen working together. Oxygen alone reacts with iron very slowly because the oxygen molecules cannot easily penetrate the metal surface. Water dissolves oxygen and carries it to the metal surface, and also conducts the electron transfer that makes the reaction possible. This is why metal in dry air barely rusts, metal underwater with no dissolved oxygen barely rusts, but metal in wet, airy conditions rusts rapidly. Salt accelerates rusting even further because salt water conducts electrons better than fresh water, speeding up the chemical reaction.

How We Protect Metal

Engineers use several strategies to prevent rust, all based on blocking water or oxygen from reaching the iron: Paint creates a physical barrier that water and oxygen cannot penetrate. Oil and grease coat the surface and repel water. Galvanizing (coating iron with zinc) provides a sacrificial layer — the zinc corrodes instead of the iron. Chrome plating creates a shiny, impermeable barrier. Stainless steel is an alloy that includes chromium, which forms its own invisible protective oxide layer. Even keeping metal dry and clean extends its life significantly.

Rust in the Real World

Rust is responsible for bridge collapses, ship sinkings, and building failures throughout history. The Statue of Liberty required a massive 1986 restoration because its internal iron framework had corroded. Ships in saltwater environments rust so quickly that they require constant maintenance — a ship that is not painted and maintained can lose structural integrity within a few years. On the other hand, some ancient iron artifacts have survived thousands of years because they were kept dry. The Iron Pillar of Delhi, India, has resisted rust for over 1,600 years due to its unique chemical composition that forms a natural protective layer.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Water Exposure and Oxygen Level as external components (environmental conditions) and Oxidation Rate, Rust Formation, and Metal Strength as internal components that respond to the external conditions.

E **Establish Relationships**

Students determine that Water Exposure positively affects Oxidation Rate, Oxygen Level positively affects Oxidation Rate, Oxidation Rate positively affects Rust Formation, Rust Formation negatively affects Metal Strength, and Water Exposure positively affects Rust Formation.

V **Visualize & Model**

Students build a five-component model showing how water and oxygen drive the oxidation reaction, which produces rust that progressively weakens the metal structure.

E **Evaluate Outcomes**

Students run four scenarios (wet/air, dry/air, wet/sealed, dry/sealed) to observe how water and oxygen combine to determine rust formation, discovering that removing either ingredient dramatically slows the process.

R **Revise & Extend**

Students add salt presence, temperature, or metal type to explore why coastal environments destroy metal faster, why warmer climates accelerate rusting, and why some metals resist corrosion.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at these two chains. They are made of the exact same metal. But one has been sitting in the rain for months. What happened to it?"*

Watch For

Pass around a rusty nail and a shiny nail (or show close-up photos). Let students feel the difference. Ask: Is the rusty nail just a different color, or has the metal itself changed?

Slide 2: Learning Objectives

3 min

Say: *"Today we are investigating one of the most common chemical reactions on Earth. This reaction costs the country billions of dollars every year!"*

Watch For

Have students read objectives aloud. Pre-teach 'chemical change' by comparing it to physical change (tearing paper vs. burning paper).

Slide 3: Big Question

3 min

Say: *"Metal is one of the strongest materials we have. So how does rain and air turn it into something you can crumble with your fingers?"*

Watch For

Turn and talk: What do you think rust actually IS? Is it just dirty metal, or is it something completely different? Write predictions.

Slide 4: LEVER Framework

2 min

Say: *"We control two environmental factors — how wet the metal is and how much oxygen reaches it. Three things happen as a result."*

Watch For

Preview all five components. Ask: Do you think water or oxygen is more important for causing rust?

Slide 5: Activity 1: Components

8 min

Say: *"Water Exposure and Oxygen Level are external — they are environmental conditions. Oxidation Rate, Rust Formation, and Metal Strength are internal — they are what happens to the metal."*

Watch For

Guide component sorting. Emphasize that Metal Strength is internal because it changes as a RESULT of the rust forming, not as a design choice.

Slide 6: Activity 2: Connections

10 min

Say: *"Both water and oxygen INCREASE oxidation rate. More oxidation means more rust. And here is the important one — more rust means LESS metal strength. The rust is eating the metal!"*

Watch For

Map relationships. Demonstrate with a real example: show how a rusty nail breaks more easily than a shiny nail.

Slide 7: Activity 3: Simulation

12 min

Say: *"Watch what happens in each scenario. The rainy, open-air bike is in trouble. But look at the dry, sealed bike — almost no rust at all!"*

Watch For

Run all four scenarios. Focus on comparing wet/air vs. wet/sealed — same water, but blocking oxygen makes a huge difference.

Slide 8: Discoveries

5 min

Say: *"Rust is a chemical change — the iron atoms combine with oxygen to form a brand new substance. And you need BOTH water and oxygen for it to happen."*

Watch For

Quick experiment: put a drop of water on a piece of steel wool. Come back in 20 minutes — students will see rust starting already! Compare to dry steel wool.

Slide 9: STEM Challenge

12 min

Say: *"A bike company needs YOUR help! Test different coatings to find the best way to protect metal from rust. You are materials scientists now!"*

Watch For

Distribute materials (steel nails, cups of water, petroleum jelly, nail polish, cooking oil, wax, aluminum foil). Groups coat nails and place in water. Check results after one week.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Water Exposure, Oxygen Level
- **INTERNAL:** Oxidation Rate, Rust Formation, Metal Strength

Relationships

- **Water Exposure to Oxidation Rate:** POSITIVE (+) — More water on the metal surface speeds up the oxidation reaction. Water acts as a catalyst, dissolving oxygen and carrying it to the iron atoms, and also conducting the electron transfer needed for the reaction. Wet metal rusts many times faster than dry metal.
- **Oxygen Level to Oxidation Rate:** POSITIVE (+) — More oxygen available means more oxygen molecules can react with iron atoms. Oxygen is one of the two essential reactants (along with water) for the rusting reaction. Higher oxygen levels provide more fuel for the chemical reaction.
- **Oxidation Rate to Rust Formation:** POSITIVE (+) — A faster oxidation rate means rust forms more quickly. When the chemical reaction is happening rapidly, iron atoms are being converted to iron oxide at a faster pace, and visible rust appears sooner and accumulates faster.
- **Rust Formation to Metal Strength:** NEGATIVE (-) — As more rust forms, metal strength decreases. Rust (iron oxide) is weaker, more brittle, and flakier than the original iron. Each layer of rust that forms replaces solid metal with a weak, porous substance. Eventually, enough rust can cause the metal to crumble and break.
- **Water Exposure to Rust Formation:** POSITIVE (+) — In addition to speeding up the oxidation rate, water directly contributes to rust formation by maintaining the wet conditions that sustain the ongoing chemical reaction. Continuously wet metal forms rust faster than metal that only gets wet occasionally.

Key Discoveries

- Rust forms fastest when BOTH water and oxygen levels are high. The chemical reaction needs both ingredients to occur
- Removing EITHER water or oxygen significantly slows rust formation. You do not need to eliminate both — reducing one dramatically helps
- Rust is a NEW SUBSTANCE (iron oxide) that is weaker and flakier than the original metal. This is a chemical change, not just a surface stain
- Protecting metal from water (keeping it dry) or from oxygen (coating it with paint, oil, or chrome) prevents rust by removing one of the necessary ingredients

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Rust is just the metal getting dirty or stained"	Rust is NOT dirt or a stain — it is a completely new substance created by a chemical reaction. When iron reacts with oxygen and water, the iron atoms rearrange and combine with oxygen atoms to form iron oxide (Fe_2O_3). This new substance has entirely different properties: it is weaker, lighter, flakier, and a different color. You cannot wash rust off because it IS the metal — the iron itself has been chemically transformed. This is the difference between a physical change (like getting mud on metal, which you can wash off) and a chemical change (like rusting, which permanently alters the substance).
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All metals rust the same way"	Only iron and iron-containing metals (like steel) form the reddish-brown substance we call rust (iron oxide). Other metals corrode differently: copper turns green (patina/verdigris), aluminum forms a thin invisible oxide layer that actually PROTECTS it from further corrosion, and gold does not corrode at all. Each metal reacts with oxygen and water in its own unique way, producing different substances. 'Rust' specifically refers to iron oxide.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Putting metal underwater will make it rust fastest because there is more water"	While water is essential for rusting, metal submerged in water with NO dissolved oxygen actually rusts VERY SLOWLY. Remember, rusting needs BOTH water AND oxygen. Deep underwater, there is less dissolved oxygen, so the reaction slows down. The FASTEST rusting happens at the waterline — where metal is exposed to both liquid water AND air at the same time. Ships rust fastest at the waterline for exactly this reason.